

Recognizing the maximum finite number of solutions of an absolute value equation

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Matthew J. Colbrook
Department of Applied Mathematics and Theoretical Physics
University of Cambridge, Cambridge, United Kingdom
m.colbrook@damtp.cam.ac.uk
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Abstract

For rational $A \in \mathbb{R}^{n \times n}$ and $b \in \mathbb{R}^n$, we give a polynomial-time test for whether $Ax + |x| = b$ has exactly 2^n distinct solutions. The test uses $n+1$ rational linear-programming feasibility problems. No nonsingularity or spectral-radius promise is required; infinite solution sets are rejected. The proof characterizes the maximum finite solution count by the geometry of a polytope with paired inequalities.

1 Statement and relation to the problem

Absolute values and inequalities are componentwise. Put $D_d = \text{diag}(d)$ and

$$P = \{y : (A + I)y \leq b, (A - I)y \leq b\}, \quad u(y) = b - Ay.$$

Thus $y \in P$ if and only if $u(y) \geq |y|$. The repository asks for polynomial-time recognition of exactly 2^n solutions, including the possibility of singular systems and infinitely many solutions in negative instances [1]. Earlier sufficient assumptions include a spectral-radius restriction [2].

Theorem 1. *The following are equivalent:*

1. $Ax + |x| = b$ has exactly 2^n distinct solutions.
2. $b > 0$, P is bounded, and $u_i(y) > 0$ for every $y \in P$ and $i = 1, \dots, n$.

2 Necessity

In the closed orthant indexed by $s \in \{-1, 1\}^n$, the solution set is

$$\{x : (A + D_s)x = b, D_s x \geq 0\}.$$

It is convex. A finite such set contains at most one point. Count incidences between solutions and closed orthants. If there are exactly 2^n solutions, each orthant contains exactly one solution, and each solution belongs to exactly one closed orthant. In particular all its coordinates are nonzero. Denote the solution by $x_s = D_s u_s$, where $u_s > 0$. The matrix $A + D_s$ is nonsingular: otherwise its nullspace would give a nonconstant line segment of solutions in the open orthant. Consequently $N_s = I + AD_s$ is nonsingular and $N_s u_s = b$.

For $d \in [-1, 1]^n$, let

$$q(d) = \det(I + AD_d),$$

and let $r_i(d)$ be the determinant obtained by replacing column i of $I + AD_d$ by b . Both polynomials are multiaffine, and r_i does not depend on d_i . Cramer's rule gives

$$r_i(s) = q(s)(u_s)_i.$$

Flipping s_i leaves the left side unchanged, so the determinants at two adjacent cube vertices have the same sign. All cube vertices are connected by such flips. Their determinant average is $q(0) = 1$, hence every $q(s)$ is positive. Every $r_i(s)$ is therefore positive as well. Multiaffine interpolation yields

$$q(d) > 0, \quad r_i(d) > 0 \quad (d \in [-1, 1]^n).$$

At $d = 0$ this implies $b_i = r_i(0) > 0$. The continuous function

$$u(d) = (I + AD_d)^{-1}b$$

is strictly positive on the entire cube.

For any $y \in P$, define $u = b - Ay \geq |y|$. Set $d_i = y_i/u_i$ when $u_i > 0$, and set $d_i = 0$ when $u_i = 0$ (which also forces $y_i = 0$). Then $y = D_d u$ and $(I + AD_d)u = b$, so $u = u(d) > 0$. Conversely, $y = D_d u(d)$ is in P for every d in the cube. Thus

$$P = \{D_d u(d) : d \in [-1, 1]^n\}.$$

This is the continuous image of a compact set, and all the asserted slack inequalities hold.

3 Sufficiency

Assume condition 2. The origin satisfies all $2n$ inequalities strictly, so P is nonempty and full dimensional. In pair i , the inequalities are $y_i \leq u_i(y)$ and $-y_i \leq u_i(y)$. They cannot both be active at a feasible point, since that would force $y_i = u_i(y) = 0$.

A vertex of a full-dimensional polytope requires n linearly independent active normals. At most one normal per pair can be active. Therefore each vertex has exactly one active inequality from each pair, and those n normals are independent. Its coordinates are nonzero, and it is an AVE solution.

We show that the sign patterns of vertices are closed under flipping any one coordinate. Start at a vertex and retain its $n - 1$ active equations from all pairs except pair i . Their common nullspace is one dimensional. Choose the direction that strictly relaxes the omitted active inequality. A sufficiently small step in this direction is feasible. Boundedness gives a finite last feasible point on the resulting ray. At this endpoint some new inequality must become active. It cannot be the opposite of any retained inequality, since two members of a pair cannot be active together. It cannot be the inequality being strictly relaxed. It must therefore be the opposite member of pair i . Its normal is independent of the retained normals, since its value changes along the ray. Hence the endpoint is a vertex with exactly sign i flipped.

A nonempty bounded polytope has a vertex. Repeated flips now produce a vertex in every open orthant. For each pattern, its active system is nonsingular, so there is at most one such vertex. Conversely, an AVE solution lies in P , is strict in sign, and satisfies one of these nonsingular active systems. It equals the corresponding vertex. There are exactly 2^n solutions, proving Theorem 1.

4 The algorithm and bit complexity

First reject if any $b_i \leq 0$. Test feasibility of

$$(A + I)r \leq 0, \quad (A - I)r \leq 0, \quad -\mathbf{1}^T Ar = 1. \quad (1)$$

Reject if feasible. Indeed, a recession vector satisfies $-Ar \geq |r|$. For every nonzero such vector, $-\mathbf{1}^T Ar \geq \|r\|_1 > 0$, so it can be scaled to satisfy the last equality. Conversely, any feasible vector in (1) is a nonzero recession vector. Since P is already nonempty, this is exactly the boundedness test.

For each i , test feasibility of

$$(A + I)y \leq b, \quad (A - I)y \leq b, \quad A_{i,:}y = b_i. \quad (2)$$

Reject if any of these n systems is feasible, and accept otherwise. On P , the final equality is precisely $u_i(y) = 0$; all u_i are nonnegative there. Theorem 1 proves correctness.

There are $n + 1$ feasibility tests, each with n variables, $2n$ inequalities, and one equality. Their rational coefficients have polynomial encoding length in the input. Polynomial-time rational linear programming [3] therefore gives a polynomial bit-complexity algorithm. This statement concerns exact rational linear programming; floating-point implementations in the accompanying tests are only diagnostic.

Remark 2. Boundedness is essential. For $n = 1$, $A = 2$, $b = 1$, one has $P = (-\infty, 1/3]$ and $1 - 2y > 0$ throughout P , but the equation has only one solution.

5 An exact example beyond the spectral-radius restriction

For

$$A = \frac{3}{5} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad b = (1, 2)^T,$$

the four solutions are

$$(10/73, 95/73)^T, \quad (-10/7, 5/7)^T, \quad (40/7, -95/7)^T, \quad (-40/13, -5/13)^T.$$

Direct substitution verifies each equation, and the four sign systems are nonsingular. Meanwhile $|A|$ has spectral radius $6/5 > 1$. This example illustrates that the theorem is not confined to the earlier spectral-radius subclass.

References

- [1] OpenProblemsInNLA, problem AV-01. <https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/intervals-and-absolute-value-equations/AV-01>.
- [2] M. Hladík, *Absolute value equations with 2^n solutions*, Optimization Letters 20 (2026), 559–575. <https://doi.org/10.1007/s11590-025-02251-z>.
- [3] N. Karmarkar, *A new polynomial-time algorithm for linear programming*, Combinatorica 4 (1984), 373–395. <https://doi.org/10.1007/BF02579150>.